EXPERIMENT- 05

Python Code:

```python
05.py > ...
def mod_inverse(a, m):
    for x in range(1, m):
        if (a * x) % m == 1:
            return x
    return -1

print("a. Limitations on b: None.")
print("b. Values of a not allowed: Even numbers and 13.")

a = int(input("Enter key a: "))
b = int(input("Enter key b: "))
plaintext = input("Enter plaintext: ").upper()

a_inv = mod_inverse(a, 26)

if a_inv == -1:
    print("Invalid value for 'a'. It must be coprime to 26.")
else:
    encrypted_text = ""
    for char in plaintext:
        if char.isalpha():
            p = ord(char) - 65
            c = (a * p + b) % 26
            encrypted_text += chr(c + 65)
        else:
            encrypted_text += char

    print("Encrypted Text:", encrypted_text)

    decrypted_text = ""
    for char in encrypted_text:
        if char.isalpha():
            c = ord(char) - 65
            p = (a_inv * (c - b)) % 26
            decrypted_text += chr(p + 65)
        else:
            decrypted_text += char

    print("Decrypted Text:", decrypted_text)
```

Output:

```
PS D:\College\Crypto\Experiments> python -u "d:\College\Crypto\Experiments\05.py"
a. Limitations on b: None.
b. Values of a not allowed: Even numbers and 13.
Enter key a: 5
Enter key b: 8
Enter plaintext: HELLO
Encrypted Text: RCLLA
Decrypted Text: HELLO
PS D:\College\Crypto\Experiments>
```